Permutations & Combinations

A Complete Lesson Plan

Grade 6 • Indian Diversity Edition

Subject	Mathematics
Grade Level	Class 6 (NCERT-aligned)
Topic	Permutations & Combinations
Duration	60–75 Minutes
Theme	Indian Festivals, Sports & Daily Life

Fun Fact: Bhaskara II, the great Indian mathematician, explored counting principles over 800 years ago — the roots of combinatorics!

1. Learning Objectives

Understand	the difference between permutations and combinations.
Apply	factorial notation to count arrangements.
Solve	real-life problems using nPr and nCr.
Connect	math concepts to Indian cultural contexts.
Develop	logical reasoning and systematic thinking skills.

2. Key Terms to Know

Permutation	An **arrangement** where **order matters**.	*Eg: ABC and CBA are counted as different permutations.*
Combination	A **selection** where **order does NOT matter**.	*Eg: Choosing Mango + Banana is the same as Banana + Mango.*
Factorial (!)	Multiply a number by every positive integer below it.	*Eg: 3! = 3 × 2 × 1 = 6 | 4! = 24 | 5! = 120*

3. Understanding Permutations

A **permutation** counts every **different ordering** of items. When you arrange books on a shelf or assign prize positions in a race, changing the order gives a new arrangement.

Formula — Permutation (nPr)

$$nPr = n! / (n - r)!$$

n = total number of items | r = items being arranged

Worked Example 1 — Diwali Diya Arrangement

Step	Working	Result
Given	Arrange 3 diyas (Red, Blue, Green) in a row	—
Formula	n! = 3!	—
Calculate	3 × 2 × 1	**6 ways**
List all	RBG, RGB, BRG, BGR, GRB, GBR	

Worked Example 2 — Arrange 4 Flags in a Row

How many ways can 4 different coloured flags be arranged in a row?

n = 4 (flags)	Formula: 4! × 3 × 2 × 1 = 24 ways	

4. Understanding Combinations

A **combination** counts selections where the **order does NOT matter**. When you pick 2 friends for a team, the pair Arjun+Priya is the same as Priya+Arjun — only one combination.

Formula — Combination (nCr)

n C r = n! / (r! × (n – r)!)

n = total items | r = items being chosen

Worked Example 3 — Festive Mithai Box

Pick 3 sweets from: Ladoo, Barfi, Gulab Jamun, Jalebi (n=4, r=3).

Step	Working	Result
Formula	4! / (3! × 1!)	—
Numerator	4! = 24	—
Denominator	3! × 1! = 6 × 1 = 6	—
Answer	24 / 6	4 ways

Worked Example 4 — Gully Cricket Team

Choose 5 players from 8 friends to form a gully cricket team (n=8, r=5).

8 C 5 = 8! / (5! × 3!)	= (8 × 7 × 6) / (3 × 2 × 1)	= 336 / 6 = 56 teams

5. Permutations vs Combinations at a Glance

Feature	PERMUTATION	COMBINATION
Order	Matters	Does NOT matter
Formula	n! / (n-r)!	n! / r!(n-r)!
Count	Always LARGER (or equal)	Always SMALLER (or equal)
Key Question	How many ways to ARRANGE?	How many ways to CHOOSE?
India Example	Positions in a relay race	Selecting a cricket team
India Example	Arranging diyas in a row	Picking sweets from a box

6. Real-Life Indian Examples

Context	Problem	Type	Answer
Diwali Decoration	Arrange 4 coloured diyas in a row	**Permutation**	**4! = 24**
Cricket	Choose 5 players from 8 friends	**Combination**	**8C5 = 56**
Mithai Box	Pick 3 sweets from 4 options	**Combination**	**4C3 = 4**
Flag Parade	Arrange 3 flags from 5 national colours	**Permutation**	**5P3 = 60**
Dosa Toppings	Choose 2 toppings from 4 options	**Combination**	**4C2 = 6**
Relay Race	Assign 1st/2nd/3rd from 6 runners	**Permutation**	**6P3 = 120**

7. Practice Problems

QA	How many ways can you arrange 4 different coloured flags in a row?	Permutation
Ans	**$4! = 4 \times 3 \times 2 \times 1 = 24$ ways**	

QB	In how many ways can you pick 3 books from a shelf of 5 books (order doesn't matter)?	Combination
Ans	**5C3 = $5! / (3! \times 2!) = 120/12 = 10$ ways**	

QC	How many ways can you arrange 3 diyas in a line for Diwali decoration?	Permutation
Ans	**$3! = 3 \times 2 \times 1 = 6$ ways**	

QD	You have 4 ladoos. In how many ways can you choose 2 to share with friends?	Combination
Ans	**4C2 = $4! / (2! \times 2!) = 24/4 = 6$ ways**	

QE	Assign 1st, 2nd, and 3rd place from 6 runners in a school race.	Permutation
Ans	**6P3 = $6! / 3! = 720/6 = 120$ ways**	

QF	A team of 4 is chosen from 7 students for a math quiz. How many possible teams?	Combination
Ans	**7C4 = $7! / (4! \times 3!) = 5040/144 = 35$ teams**	

8. Quick Quiz — Test Your Understanding

Circle or underline the correct answer for each question below.

Q1. True or False: In permutations, the order of items matters.

A) True
B) False

Answer: A) True — Order always matters in permutations.

Q2. Choosing 2 toppings for your dosa from 4 options is a:

A) Permutation
B) Combination

Answer: B) Combination — order of toppings doesn't matter!

Q3. What is 4! (4 factorial)?

A) 12
B) 24
C) 16

Answer: B) 24 ($4 \times 3 \times 2 \times 1 = 24$)

Q4. Arranging 3 children in a row for a photo is a:

A) Combination
B) Permutation

Answer: B) Permutation — different orders give different photos.

Q5. How many ways can you choose 2 friends from 5 for a project team?

A) 10
B) 20
C) 15

Answer: A) 10 ($5C2 = 5!/(2!\times3!) = 10$)

9. Lesson Summary

PERMUTATIONS	COMBINATIONS
Order MATTERS	Order does NOT matter
Formula: nPr = n! / (n-r)!	Formula: nCr = n! / r!(n-r)!
Larger count	Smaller count
Arranging books, diyas, runners	Choosing teams, sweets, books
Eg: 3P2 = 6	Eg: 3C2 = 3

10. Memory Tips & Tricks

1	P for Position — use Permutation when positions (1st, 2nd, 3rd) matter.
2	C for Crew — use Combination when picking a group or crew.
3	Permutation ≥ Combination always! nPr = nCr × r!
4	To simplify nCr, cancel common factors top and bottom before multiplying.
5	Always ask: "Does swapping two items change the result?" Yes=P, No=C.

Bonus Maths Fact! The great Indian mathematician **Bhaskara II** explored counting principles in his 12th-century work *Lilavati* — one of the earliest systematic treatments of permutations and combinations. Math connects us across centuries!